

Bole Ma

ma929@purdue.edu | [Website](#) | [LinkedIn](#) | [Google Scholar](#)

Research interests: Computer vision, geometric perception, and computational imaging.
Seeking Summer 2027 research and engineering internships.

Education

Purdue University Ph.D. in Electrical and Computer Engineering Advisor: Prof. Stanley H. Chan, Intelligent Imaging Lab	Sep 2024–Present; Expected May 2029 GPA: 4.00/4.00
Shanghai Jiao Tong University M.S. in Electronic Information and Electrical Engineering	Sep 2021–Jun 2024 GPA: 3.64/4.00
Shanghai Jiao Tong University B.S. in Electronic Information and Electrical Engineering	Sep 2017–Jun 2021 GPA: 3.79/4.30

Technical Skills

Programming	Python, MATLAB, C++ (basic)
Deep Learning	PyTorch, neural-network training
Research Computing	Linux, Slurm, AI-assisted research workflows
Vision & Imaging	Geometric modeling, monocular depth estimation, camera calibration, optical system design and alignment
Experimental Methods	Dataset curation, experimental design, wavefront reconstruction

Research Experience

Geometry-Guided Monocular Metric Depth Estimation <i>Computer Vision · Purdue University</i>	Jan 2026–Present
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Developing an interpretable monocular metric depth framework for robotic and UAV perception, integrating explicit camera geometry with learned scene representations.

- Designed and implemented an interpretable depth-estimation framework integrating explicit camera geometry with learned scene representations.
- Built and ran the training pipeline, enabling metric depth prediction from monocular RGB images.
- Conducted controlled intervention experiments to assess how intermediate predictions influence depth outputs and guide ongoing model refinement.
- Curated task-specific training subsets from approximately 4.72 million TartanAir v2 frames, balancing scene coverage and camera-geometry diversity under a single-A100 compute budget.

DARPA CIDAR Challenge <i>Student Project Lead · Purdue University</i>	Jan 2025–Jun 2026 DARPA · Purdue News
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Passive image-based ranging for high-accuracy, low-latency distance estimation.

- Led data organization, analysis, and feasibility assessment of infrared-absorption, stereo-vision, and monocular depth-estimation approaches, guiding the team's method selection.
- Collected approximately 1,000 UAV images, calibrated and organized the dataset, and trained a compact U-Net prototype; identified domain shift as a limitation of training solely on locally collected data.
- Developed a camera-geometry-based approach for passive image-based distance estimation, following feasibility assessment and neural-network prototyping.
- Helped Purdue place among the five semifinal awardees; the team received \$250,000 across white-paper and semifinal awards.

Pupil Design for Computational Wavefront Estimation

Jan 2026–Apr 2026

Computational Imaging · Purdue University

Investigating asymmetric pupil designs to resolve ambiguity in wavefront phase recovery from point spread functions (PSFs).

- Designed, built, and refined an optical testbed, including component selection, optical alignment, and troubleshooting.
- Captured 3,000+ experimental PSF images, of which 2,500 were selected for subsequent neural-network fine-tuning and wavefront reconstruction.

Selected Prior Research**Complex Wavefront Retrieval with a Higher Space–Bandwidth Product**

Sep 2022–Jun 2023

Computational Imaging · Shanghai Jiao Tong University

Master’s Research

Recovering complex wavefronts—including amplitude and phase—from diffraction intensity images while preserving field of view and spatial resolution.

- Built and trained an imaging-process-matched neural network that processes full-size diffraction patterns to reconstruct complex wavefronts with a larger field of view and higher spatial resolution.
- Designed the optical experimental setup, collected data, and validated reconstruction through simulations and physical experiments; wrote the first-author paper published in *Optics Letters* (2023).

Selected Publications

1. Weijian Zhang, Prateek Chennuri, Hashan K. Weerasooriya, **Bole Ma**, Stanley H. Chan. **Markov-Renewal Single-Photon LiDAR Simulator**. *European Conference on Computer Vision (ECCV)*, 2026.
2. Yu Yuan, Xijun Wang, Tharindu Wickremasinghe, Zeeshan Nadir, **Bole Ma**, Stanley H. Chan. **NewtonGen: Physics-Consistent and Controllable Text-to-Video Generation via Neural Newtonian Dynamics**. *International Conference on Learning Representations (ICLR)*, 2026.
3. Ali Almualllem, Nicholas Chimitt, **Bole Ma**, Qi Guo, Stanley H. Chan. **Pupil Design for Computational Wavefront Estimation**. *arXiv preprint*, 2026.
4. Xijun Wang, Xingguang Zhang, Yu Yuan, **Bole Ma**, Prateek Chennuri, Stanley H. Chan. **Generative Personalized Blind Face Restoration Enhanced by Physical Identity**. *IEEE International Conference on Image Processing (ICIP)*, 2025.
5. **Bole Ma**, Chuxuan Huang, Sibing Hou, Haiyu Zhang, Manhua Liu, Hao Yan. **Imaging process matched neural network for complex wavefront retrieval with a higher space–bandwidth product**. *Optics Letters*, 48(20), 5399–5402, 2023. **First author**.

Honors and Awards

- **DARPA CIDAR Challenge** — one of five semifinal awardees; Purdue team received \$250,000 across white-paper and semifinal awards, 2026.
- **Outstanding Graduate of Shanghai Municipality** — Top 5%.
- **Instrument 78 Motivation Fund**, Shanghai Jiao Tong University — Rank 1/52.
- **Endress+Hauser Scholarship**, Shanghai Jiao Tong University — Top 5%.
- **B-Grade Outstanding Scholarship**, Shanghai Jiao Tong University — Top 10%.